Chapter 4

Infection Control

"They certainly give very strange names to diseases…"

—Plato

Exposure Risks and Effect of Infections on Dentistry

Increases in serious transmissible diseases over the last few decades have created global concern and have affected the treatment approach of all health care practitioners. The goal is to ensure compliance with standard precautions and other methods to minimize infection risks.[1]

Air-borne Contamination

A high-speed hand-piece is capable of creating air-borne contaminants from bacterial residents in the dental unit water spray system and from microbial contaminants from saliva, tissues, blood, plaque and fine debris cut from carious teeth (Fig. 4.1). With respect to size, these air-borne contaminants exist in the form of spatter, mists and aerosols.

Aerosols

Aerosols consist of invisible particles ranging from *5* µm to *approximately 50* µm that can remain suspended in the air and breathed for hours.[2] Aerosols and larger particles may carry agents of any respiratory infection carried by the patient.

Mists

Mists that become visible in a beam of light consist of droplets estimated to be *around 50* µm. Heavy mists tend to settle gradually from the air after 5–15 min.[3] Aerosols and mists produced by the cough of a patient with unrecognized active pulmonary or pharyngeal tuberculosis are likely to transmit the infection.[4]

Spatter

Spatter consists of particles generally *larger than 50* µm and even visible splashes. Spatter has a distinct trajectory, usually falling within 3 ft of the patient's mouth, having the potential for coating the face and outer garments of the attending personnel.[5] Spatter or splashing of mucosa is considered a potential route of infection for dental personnel by blood-borne pathogens.[4,6,7]

Clinical Notes

- Barrier protection of personnel using masks, protective eyewear, gloves and gowns is now a *standard requirement* for dental procedures.
- A pretreatment mouthrinse, *rubber dam* and high-velocity air evacuation also can reduce microbial exposure.[5,7]
- To help reduce exposure to air-borne particles capable of transmitting respiratory infections, adequate *air circulation* should be maintained.[6]

Direct and Indirect Contamination

Direct contamination occurs during *direct contact* with bodily fluids, and this is a major exposure concern for dental personnel.

Indirect contamination occurs due to saliva-contaminated hands; the dentist and the assistant could repeatedly contact or handle unprotected operatory surfaces during treatment. Soiled surfaces that are poorly cleaned provide another source of gross environmental contamination and thus potential cross-contamination of personnel and patients.[8,9]

Items or areas still contaminated after cleaning may include hand-pieces; air-water syringe handles; control switches on the patient's chair; seat edges and rests of the dentist's and assistant's chairs; knobs; cabinet, drawer and operatory tray handles; room light switches and operatory telephones.



Figure 4.1 Dentistry as it may have been practiced in the past. Rotary instrumentation can expose personnel to heavy spatter of more than 50-micron particles and mists. Aerosol particles of less than 5 micron remain suspended and can reach the alveoli if not stopped by a barrier. Air purification is a growing concern.
(Courtesy: Laminaire Corporation, Palm Beach Gardens, FL, USA)

Clinical Notes

Barrier protection of personnel and equipment (Fig. 4.2), instrument sterilization and methods of avoiding direct contact with various surfaces are mandatory.[10–24]

Personal Barrier Protection

Instructions for Handwashing

At the beginning of a routine treatment period, the clinician should remove his or her wristwatch, jewellery and rings then wash hands with a suitable cleanser.

Clinical Notes

- Gloves must not be used for more than one patient.
- Inexpensive, disposable, well-fitting treatment gloves are available for chair-side use. *Gloves must not be washed*. Washing reduces the integrity of the glove, leaving personnel more vulnerable. Instead of attempting to wash gloved hands before opening drawers or handling items adjacent to the operatory, tongs, a paper towel or a food handler's overglove should be used to prevent contamination.



Figure 4.2 In current dental practice, personal protective equipment (PPE) provides barriers against spatter and aerosols during patient treatments.
(From Bird DL, Robinson DS: *Modern Dental Assisting,* ed 10, St. Louis, MO, 2012, Saunders.)

Additionally, proper use of *alcohol rubs* is effective against pathogens and less drying to the hands.

Gloves

Regulations specify that all clinical personnel *must wear* treatment gloves during all treatment procedures. After each appointment, or whenever a leak is detected, gloves are removed, hands are washed and fresh gloves are donned (Figs. 4.3 and 4.4).



Figure 4.3 To remove a contaminated glove, pinch the palm side of the outer cuff surface with the gloved fingers of the other hand. Pull off the glove, inverting it. Both gloves can be removed simultaneously in this manner. Alternatively, after removing one, insert bare fingers under the cuff to grasp and pull off the remaining glove. Discard gloves safely.



Figure 4.4 To wash hands after removing treatment gloves, operate the pump as shown with the clean underside of a wrist. Also, operate faucet handles the same way to avoid contamination or use foot controls. Never touch the handles with contaminated gloves.



Figure 4.5 Remove the mask as shown. Grasp the mask ties or elastic band behind the head instead of grasping the contaminated mask. Before treatment, put on mask and eyewear before washing and gloving hands. After treatment, remove gloves and then eyewear and mask and wash hands.

Protective Eyewear, Masks and Hair Protection

Protective Eyewear

They may consist of goggles or glasses with solid side-shields. When eyewear or shields are removed, they should be cleaned and disinfected.

Clinical Notes

- To save time, clean replacement eyewear should be readily available while used eyewear is being disinfected.
- Antifog solution for eyewear is clinically recommended.

Masks

A mask should be worn to protect against aerosols. Face shields are appropriate for protection against heavy spatter, but a mask still is required to protect against aerosols that drift behind the shield.[5,14]

Clinical Notes

- The clinician should grasp the mask only by the string or band at the sides or back of the head to remove it (Fig. 4.5).
- The mask should be changed between every patient or whenever it becomes moist or visibly soiled.
- Touching masks and eyewear during treatments should be avoided to prevent cross-contamination.
- To protect against aerosols, the edges of the rectangular mask should be pressed close around the bridge of the nose and face.

Hair Protection

Operatory personnel should keep their hair out of the treatment field. Hair can trap heavy contamination that, if not washed away, can be rubbed back from a pillow onto the face at night.

Clinical Notes

Hair must be protected with a surgical cap.

Protective Overgarments

Operatory clothing is heavily spattered with invisible saliva and traces of blood throughout the day. HBV and many other microbes can live on dry materials for 1 or more days.[3,10,25,26] An overgarment must protect clothing as well as skin.

Clinical Notes

- A simple, lightweight garment that covers the arms and chest up to the neck and the lap when seated may provide adequate protection.
- Sleeves with knit cuffs that tuck under the gloves are preferable. Cloth made of cotton or cotton–synthetic fibre similar to isolation garment material may be thick enough to protect skin and street clothing from spatter during most dental treatments.
- Used overgarments should be only minimally handled and laundered or disposed of properly.

Disposal of Clinical Waste

Infected blood and other liquid clinical waste, except mercury, silver or other heavy metal chemicals, generally can be poured down a sanitary sewer or drain designated for that purpose. Application of aseptic precautions and cleaning and disinfection of the basin around the drain must be performed. Contaminated materials such as used masks, gloves, blood-soaked or saliva-soaked sponges and blood-soaked or saliva-soaked cotton rolls must be discarded safely. OSHA Occupational Safety and Health Administration regulations presented previously describe the rules and required labels with regard to disposal of sharps and soft waste. OSHA labelling requirements may differ from local protection agency requirements. As pathologic waste, excised tissues require separate disposal and should not be discarded into the trash.

Care must be exercised in bagging medical waste so that injury or direct contact with liquids does not occur, as HIV and HBV can survive beyond a few days in wet blood. Separating needles and sharps into hard-walled, leak-proof and sealable containers and out of soft trash has been shown to provide adequate safety. Nevertheless, local laws governing waste disposal range from the recommendations of the Centers for Disease Control and Prevention (CDC) to regulations requiring stricter management and tracking of waste disposal, usually at an added expense.[1,7] Local city, county and state regulations should be consulted.

Needle Disposal

The goals with regard to needle disposal are

I. disposing of needles in a hard-walled, leak-proof and sealable container, which has the OSHA biohazard label;
II. locating the needle-disposal container in the operatory close to where the needle will be used and
III. avoiding carrying unsheathed contaminated needles or containers in a manner that could endanger others or would allow the needles to be accidentally spilled.[14]

Local regulations for the disposal of the container should be followed.

Precautions to Avoid Injury Exposure

Pointed instruments without a hollow lumen have minimal capacity to transmit infected blood into a puncture site. The same principles that apply to needles should be reasonably translated and applied, however, to used burs, wires and sharp instruments from the operatory.

- Great care should be used in passing instruments and syringes with unsheathed needles to another individual.
- Sharp and curved ends should be turned away from the recipient's hand.
- Two-handed resheathing of needles is not permitted.
- A needle sheath holder or other safety device or technique should be used for the operator to resheath the needle with only one hand.[27]
- Burs should be removed from hand-pieces when the procedure is finished; if left in the hand-piece in a hanger, the bur should be pointed away from the hands and body.
- Hanging hand-pieces upside down in some types of hangers can angle the bur away from the operator.

Overview of Aseptic Techniques

The concept of asepsis is to prevent cross-contamination—all items that are touched with saliva-coated hands must be rendered free of contamination before beginning treatment on the next patient. These contaminated items can be discarded; protected by disposable covers or removed, cleaned and sterilized. The clinician should not directly touch what he or she does not want to contaminate. A few simple rules that help avoid wasting costly time and effort between patients appointments are given in Box 4.1.

Disinfectants

Regarding disinfection, two principles should be remembered:

I. Disinfection cannot occur until fresh disinfectant is reapplied to a thoroughly cleaned surface.[17,22]
II. Disinfection does not sterilize.[28–30]

Clinical Notes

- Disinfectants containing *70%–79% ethyl alcohol* are considered the most effective disinfectants on cleaned surfaces.[12,20,29]
- Sterilants used for high-level disinfection of items for reuse are *glutaraldehydes* at 2%–3% concentrations.

Box 4.1 Rules to prevent cross-contamination

During each appointment:
I. Remember, whatever is touched is contaminated.
II. Directly touch only what has to be touched (anticipate your needs).
III. Use one of the following to control contamination:
  i. Clean and sterilize dental instruments.
  ii. Protect surfaces and equipment that are not sterilized with disposable, single-use covers (barriers). Discard them after every appointment. Use disposable covers on portable items (e.g. curing-lamp handles, amalgam mixers and plastic air-water syringe tips).
  iii. Use a paper towel, tongs or plastic bag over gloves to handle equipment briefly or to open cabinets and drawers to get things that were not anticipated during setup.
  iv. Scrub and disinfect noncritical surfaces as well as possible. These include any countertops that cannot be covered (and may collect aerosols or spatter) or things that may be accidentally touched, such as room door handles and light switches. With practice, these areas should not become contaminated.

Procedures, Materials and Devices for Cleaning Instruments before Sterilization

Following are step-by-step standard operating procedures for the preparation of the dental chair, dental unit and instruments between appointments.

I. With gloved hands after the last treatment, remove and invert the chair back cover, discard cotton rolls and other disposable materials into the cover and discard the cover into the operatory trash bin. Remove and discard gloves aseptically.
II. Wash hands with antiseptic hand soap, rinse and dry or use an accepted alcohol hand rub.
III. Place three paper towels on the seat of the dental chair for later placement of air-water syringe and ends of suction hoses. Don nitrile latex utility gloves.
IV. With the used suction tip, clean saliva and debris from the cuspidor trap, if present. Discard the disposable suction tip into the operatory trash bin.
V. Remove (unscrew) the resheathed needle from the anesthetic syringe, and discard it with all other sharp disposable items in a sharps container. Using a Stickshield is advised. Remove the anesthetic cartridge before removing the needle to decrease the risk of an occupational needlestick injury. Handling needles without using a protective one-handed capping device and gathering instruments without heavy protective gloves account for most injury exposure incidents.
VI. Place any loose sharp instruments and instrument cassettes into a perforated metal basket, and then lower the basket into the disinfectant solution in a covered hard-walled pan. Return the hand-pieces and the pan of instruments to the cleanup area. Using the handles provided, remove the basket of instruments, rinse and place into the ultrasonic cleaner.
VII. Before handling disinfectant-dispensing bottles, wash the utility gloves (on hands) with antiseptic scrub, rinse and dry.
VIII. Spray any used bottles, containers and tubes with disinfectant, and wipe with a paper towel. Spray again, and leave the items damp with disinfectant as they are put away. Spraying in this manner has been found to be effective.[29] Disinfectant wipes that are available in the marketplace can be used, if any concern exists about breathing in irritating or possibly harmful aerosols from spray disinfectants.
IX. Remove the air-water syringe (now minus its removable tip) and suction hoses from the hangers on the control unit. Remove the plastic covers from the hose ends and discard. Lay the air-water syringe and suction hose ends on the paper towels previously placed on the dental chair.
X. Invert, remove and discard the plastic drapes from the control unit remove and discard the protective covers from lamp handles and the surface covering from the side table. These disposables may be placed into the large bag and removed from the control unit.
XI. For any controls and switches that were not covered, use a disinfectant wipe to wipe the lamp switch and controls that were contaminated. Do not spray control switches. Wipe any contaminated surfaces not previously covered, including the side table, arms of dental chair, contaminated drawer handles, radiographic viewbox switch and paper towel dispenser. Discard the used disinfectant wipe.
XII. Using a second disinfectant wipe, rewet these items and leave them wet.
XIII. Spray the outside and inside of the cuspidor, if present, with disinfectant. Use two paper towels to prevent your gloves from contacting the cuspidor while first wiping the outside and then the inside of the cuspidor. Discard the towels. Wipe any overspray of disinfectant from the operatory floor. Discard the towels in the trash bin. Disinfectant wipes could also be used.
XIV. Spray any contaminated faucet handles, sink countertops and trash disposal openings with disinfectant, and wipe dry with paper towel. Discard the towel, and re-spray the areas with disinfectant and leave them damp. Disinfectant wipes could also be used.
XV. Wash the utility gloves (still on hands) with a strong antiseptic hand scrub or disinfectant cleaner, rinse thoroughly and dry them with paper towels. Discard the towels into the trash bin. Remove the utility gloves, and re-hang them in the operatory. Wash hands. Contaminated utility gloves can be cleaned and disinfected. Nitrile latex gloves can be autoclaved.

The instruments that touch mucosa or penetrate tissues must be cleaned and sterilized before reuse (Box 4.2).[7,16]

Ultrasonic Cleaners and Solutions

Ultrasonic cleaning is the *safest* and *most efficient way* to clean sharp instruments (see Fig. 4.6). Ultrasonic cleaning can be *9 times more effective* than hand cleaning if the ultrasonic device functions properly and is used as directed by the manufacturer.[31] An ultrasonic cleaning device should provide fast and thorough cleaning without damage to instruments; have a lid, well-designed basket and an audible timer.

Procedures for ultrasonic cleaning are as follows:

I. Observe operating precautions.
II. Operate the tank at *one-half to three-fourths full* of cleaning solution at all times. Use only cleaning solutions recommended by the manufacturer of the ultrasonic device. Change solutions, as directed. An antimicrobial cleaning solution is preferable.
III. Operate the ultrasonic cleaner for *5 min* or *longer*, as directed by the manufacturer, to achieve optimal cleaning, possibly 1 min per instrument.

Box 4.2 Do's and don'ts of instrument recycling

Do:

Wear protective puncture-resistant gloves to handle used instruments.

Keep instruments wet in an antibacterial solution before cleaning.

Use an ultrasonic cleaning device.

Test and maintain the ultrasonic device periodically.

Use good-quality sterilizer equipment.

Read the operator's manual, and follow operation instructions for the sterilizer.

Have sterilizers annually inspected regarding gaskets, timer, valves and temperature and pressure gauges.

Use proper water or chemicals to operate, clean and maintain sterilizer.

Place only dry instruments in the sterilizer.

Use a wrap that will be penetrated by the steam or gas used.

Load the sterilizer loosely; leave air space between large packs.

Read the sterilizer temperature and pressure gauges daily.

Use the complete sterilizer monitoring system outlined; use indicators daily and spore tests weekly.

Keep a record of daily indicators and spore tests.

Don't:

Place wet instruments into any type of sterilizer unless so instructed.

Overwrap cloth packs or use impermeable wraps for steam or chemical vapour pressure sterilization.

Use closed, non-perforated trays, foil, canisters or other sealed containers in gas or steam sterilizers.

Overload or cram packs together in the sterilizer.

Decrease the required time for sterilization.

Add instruments to a sterilizer without restarting the cycle.

Sterilize viability control strips supplied with spore tests.

IV. Remove coatings such as plaster, wax, cement and impression material with an appropriate solvent cleaner, and place the instrument(s) and/or impression tray(s) in a beaker in the ultrasonic device.

Instrument Containment

Various kinds of instrument trays and cassettes (Fig. 4.7) are manufactured to contain the instruments at chair-side, and they can be placed in an ultrasonic cleaner, rinsed and packaged ready for sterilization. Cassettes provide convenience, safety in handling and cleaning batches of instruments, and maintenance of instrument organization for efficient use.



Figure 4.6 An ultrasonic cleaner.
(Courtesy: Midmark Corp, Versailles, OH, USA.)

Sterilization

Sterilization is defined as killing all forms of life, including the most heat-resistant forms, that is, bacterial spores. Since most instruments contact mucosa or penetrate oral tissues, it is essential that contaminated reusable instruments be cleaned and sterilized thoroughly by using accepted methods that can be tested and monitored routinely.[7,16,17]

The four accepted methods of sterilization are as follows:

I. Steam pressure sterilization (autoclave)
II. Chemical vapour pressure sterilization (chemiclave)
III. Dry heat sterilization (dryclave)
IV. Ethylene oxide (ETOX) sterilization.

Each method and each commercial modification has specific requirements with regard to timing, temperature, suitable packaging of materials and kinds of items and materials that can be sterilized safely and effectively.[27,32]

A sterilizer is used in every day of practice. Therefore, reliable sterilization equipment of proper size and cycle time compatible with needs of the practice should be chosen. Patient load, turnaround time for instrument reuse, size of instrument inventory, instrument variety and instrument

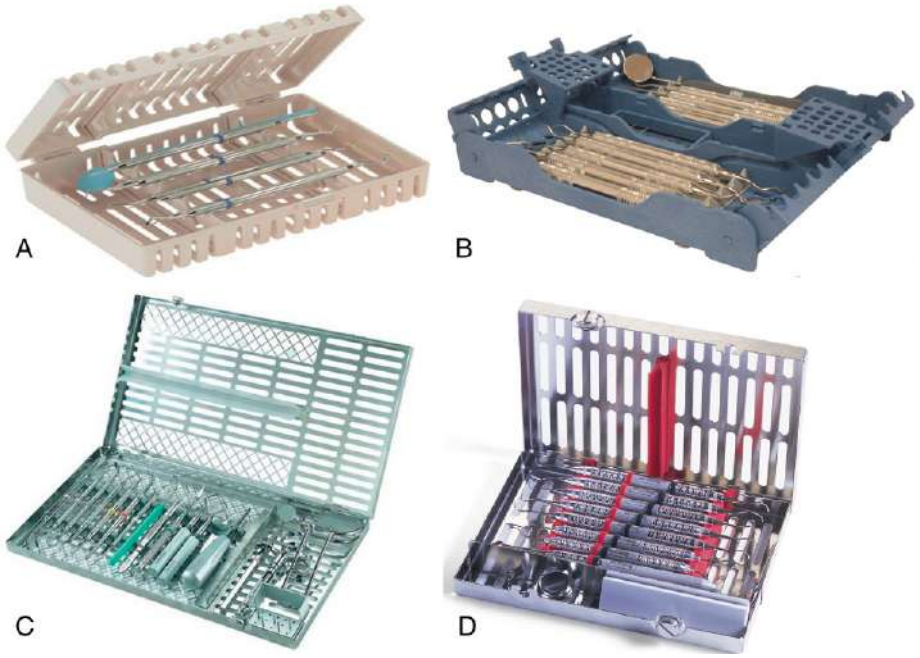


Figure 4.7 Examples of cassettes designed to hold instruments while they are cleaned and sterilized. (A) Steri System Cassette. (B) E-Z Jett Cassette. (C) Osung Dental Instrument Cassette. (D) IMS Signature Series Cassette.
(Part A, Courtesy: Dux Dental, Oxnard, CA, USA; part B, Courtesy: Zirc Company, Buffalo, MN, USA; part C, Courtesy: Brite Source USA, Chicago, IL, USA; part D, Courtesy: Hu-Friedy Mfg. Co., LLC, Chicago, IL, USA.)

quality all must be balanced against the type and size of sterilizer selected.

I. Steam Pressure Sterilization (Autoclaving)

Sterilization with steam under pressure is performed in a steam autoclave (Fig. 4.8).

Advantages of Autoclaving

- Autoclaving is the most rapid and effective method for sterilizing *cloth surgical packs* and *towel packs* (other methods are not suitable for processing cloth packs).
- Automated models are available, although they must be evaluated with a biologic spore test monitoring system.

Disadvantages of Autoclaving

- Items sensitive to the elevated temperature cannot be autoclaved.
- Autoclaving tends to *rust* carbon steel instruments and burs.
- Steam seems to *corrode* the steel neck and shank portions of some diamond instruments and carbide burs.



Figure 4.8 A steam pressure sterilizer (autoclave).
(Courtesy: Midmark Corp, Versailles, OH, USA.)

Clinical Notes

- For a light load of instruments, the time required at *250°F (121°C) is a minimum of 15 min at 15 lb of pressure* (time required for the sterilizer to reach the correct temperature is not included).
- Time for wrapped instruments can be reduced to *7 min* if the temperature is increased to approximately *273°F (134°C) to give 30 lb of pressure.*
- Instrument pans or other impermeable instrument containers must be *left open* so that steam can enter.
- Except for containers of solutions, all metal items must be dry.
- Sterilization must be tested routinely (see section on monitors of sterilization).[7,16]
- For autoclave sterilization, burs can be protected by keeping them submerged in a small amount of *2% sodium nitrite solution.*[11,15]

II. Chemical Vapour Pressure Sterilization (Chemiclaving)

They are similar to steam sterilizers but sterilization by chemical vapour under pressure is performed in a *chemiclave* (MDT Biologic Co, Rancho Dominguez, CA, USA).

Clinical Notes

- Chemical vapour pressure sterilizers operate at *270°F (131°C)* and *20 lb of pressure* and have a cycle time of approximately *30 min.*
- They must be used with a prescribed chemical. Newer models seem to handle *aldehyde vapours* well.

Advantage of Chemiclaving

Carbon steel and other corrosion-sensitive burs, instruments and pliers are said to be sterilized without rust or corrosion.

Disadvantages of Chemiclaving

- Items sensitive to the elevated temperature are damaged.
- Instruments must be lightly packaged in bags obtained from the sterilizer manufacturer.
- Towels and heavy cloth wrappings of surgical instruments may not be penetrated to provide sterilization.
- Biologic spore test monitoring strips need to be used routinely to confirm heat penetration of heavy packs before using them (see the section on monitors of sterilization).
- Only fluid purchased from the sterilizer manufacturer can be used.
- Only dry instruments should be loaded, and the door gasket should be checked for leaks to avoid frequent sterilization monitoring failures.

III. Dry Heat Sterilization

Conventional Dry Heat Ovens

Dry heat sterilization is readily achieved at temperatures greater than 320°F (>160°C).[18]

Clinical Notes

- Packs of instruments must be placed at least *1 cm apart* to allow heated air to circulate. Individual instruments must be heated at *320°F (160°C) for 30 min* to achieve sterilization.[15]
- Increasing the *total time by 50%* as a safety factor is recommended.
- Total time required also depends on the efficiency of the oven based on its size, the size of the load and how instruments are packaged.
- Approximately *60–90 min* may be required to sterilize a medium load of lightly wrapped instruments in an oven set at a range from 335°F (168°C) to 345°F (174°C).
- Without careful calibration, more sterilization failures are obtained with gravity convection dry heat ovens than any other type of sterilizer.

Short-cycle, High-temperature Dry Heat Ovens

A rapid high-temperature process that uses a forced-draft sterilization chamber (a mechanical convection sterilization chamber that circulates heated air with a fan or blower) is available.

Clinical Notes

- It reduces total sterilization time to *6 min* for unwrapped instruments and *12 min* for wrapped instruments (Fig. 4.9).
- These short-cycle, high-temperature dry heat sterilizers operate at *375°F (190°C).* Careful calibration with a pyrometer to ensure that instruments reach and maintain sterilization temperatures is imperative.

Advantages of dry heat sterilization

- Carbon steel instruments and burs *do not rust, corrode* or *lose their temper* or cutting edges if they are well dried before processing.
- Industrial forced-draft hot air ovens usually provide a larger capacity at a *reasonable price.*
- Rapid cycles are possible at high temperatures.



Figure 4.9 COX Rapid Heat brand rapid heat transfer dry heat sterilizer.
(Courtesy: CPAC Equipment Inc., Leicester, NY.)

Disadvantages of dry heat sterilization

- High temperatures may damage more heat-sensitive items such as rubber or plastic goods.
- Sterilization cycles are prolonged at lower temperatures.
- Heavy loads of instruments, crowding of packs and heavy wrapping easily prevent sterilization.
- Cycles are not automatically timed on some models.
- Inaccurate calibration, lack of attention to proper settings and adding instruments without restarting the timing are other common sources of error.

IV. ETOX Sterilization

ETOX sterilization is the best method for sterilizing *complex instruments* and *delicate materials*.

Clinical Notes

- ETOX gas is the best way to sterilize rotary handpieces.
- Automatic devices sterilize items in several hours and operate at elevated temperatures well below 100°C.
- Less expensive devices operate overnight to produce sterilization at room temperature (Fig. 4.10).
- Porous and plastic materials absorb gas and require *aeration for 24 h* or *more* before it is safe for them to contact skin or tissues.
- Units with large chamber sizes hold more instruments or packs per cycle; however, they are expensive.

Boiling Water

Boiling instruments in water does not kill spores and cannot sterilize instruments.



Figure 4.10 Room temperature ethylene oxide sterilizer.
(Courtesy: Anderson Products Inc., Haw River, NC, USA.)

Clinical Notes

- Boiling is a method of high-level disinfection that has been used *when actual sterilization cannot be achieved* (e.g. in case of a sterilizer breakdown).[28]
- Heat can reach and kill blood-borne pathogens; however, in places that liquid sterilants and disinfectants used at room temperature cannot reach.
- Well-cleaned items must be completely submerged and allowed to boil at *98–100°C (at sea level) for 10 min.*
- Great care must be exercised to ensure that instruments remain covered with boiling water the entire time.

New Methods of Sterilization

Various new methods of sterilization are under investigation and development.

I. The *microwave oven* has major limitations for sterilizing metal items, including damage of the machine caused by the metal and the inability to reach all sides of the instruments. Research efforts to overcome such limitations are ongoing.
II. *UV light* is not highly effective against RNA viruses such as HIV and is not effective against bacterial spores.[33,34] Incomplete exposures of all surfaces and poor penetration of oil and debris are other limitations. UV irradiation may be useful for sanitizing room air to help control tuberculosis bacteria.

One valuable guide to whether a commercial device is an effective sterilizer is determining whether the FDA would find it equivalent to other effective and proven devices now in common use. Before purchasing any medical device in question, the clinician should require the manufacturer to provide documentation of FDA premarketing review.

Liquid Sterilants and High-level Disinfectants

- Liquid sterilants can kill bacterial spores in 6–10 h. Sterilants used for high-level disinfection of items for reuse are *glutaraldehydes at 2%–3% concentrations*.
- Glutaraldehydes are irritating, are sensitizing to skin and respiratory passages and can be toxic as indicated in manufacturers' safety data sheets. Trays should be kept tightly covered in a well-vented area.
- Despite reuse claims of several weeks' duration, studies have shown that disinfectants in heavy use often lost activity during the second week.
- Organic matter and oxidation reduce the activity of reused disinfectant baths. Placing wet items into disinfectant trays dilutes the solution. Use of 2% or greater glutaraldehyde solutions to wipe counters or equipment (e.g. dental unit and chair) should be avoided.
- Most glutaraldehydes require 20 min to kill tuberculosis bacteria, in contrast to some synthetic phenol complexes and alcohols, which act in 10 min or less and are much less toxic.
- High-level disinfection is used mainly for plastic items that enter the mouth and that cannot withstand heat sterilization.
- Disinfection for 20–90 min in glutaraldehyde germicides is inappropriate for instruments used in the mouth. Most require 6 or more hours for sterilization.
- Plastic cheek retractors, photographic mirrors and similar heat-sensitive devices should be replaced with metal types that can be heat sterilized.
- Liquid sterilants cannot process pre-packaged instruments or be completely monitored with biologic indicators.

Clinical Notes

- According to the CDC, instruments that penetrate tissues or contact mucosa are termed critical or semicritical and require cleaning and heat or gas sterilization before reuse.[1,4]
- Periodontal, restorative and endodontic instruments are readily processed by autoclave or chemical vapour pressure sterilization.
- Carbon steel instruments and burs, if dried well before sterilizing, are best sterilized by dry heat and chemical vapour pressure sterilizers because these methods reduce the risk of rust.
- Prophy cups should be discarded and never disinfected for reuse.
- Used anaesthesia carpules and anaesthesia needles must be discarded after a patient appointment and never be disinfected or heat sterilized for reuse.

Monitors of Sterilization

Effective instrument sterilization is ensured by routine monitoring of instrument sterilization, which has become a standard of care. Sterilization monitoring has five components: (i) mechanical monitoring, (ii) chemical indicator strip in each pack, (iii) external sterilization indicator on the outside of each pack, (iv) weekly biologic spore test and (v) documentation log.

I. Mechanical Monitoring

Each sterilized load must be mechanically monitored to document time, temperature and pressure. Otherwise, the clinician manually observes the maximum temperature and pressure and documents the data in a log.

II. Chemical Indicator Strips

Chemical indicator strips provide an *inexpensive, qualitative monitor* of sterilizer function, operation and heat penetration into packs. The clinician places one of the inexpensive colour-change indicator strips into every pack. *Chemicals on the strip change colour slowly, relative to the temperature reached in the pack.* The strip is, however, not an accurate measure of sterilization time and temperature exposure.

III. External Sterilization Indicators

External sterilization indicators, including *tapes* and *bags*, are marked with heat-sensitive dyes that change colour easily on exposure to heat, pressure or sterilization chemicals. Such heat-sensitive markers are important to *identify* and *distinguish the packs that have been in the sterilizer from those that have not*. Used alone, these indicators are not an adequate measure of sterilization conditions.

IV. Biologic Monitoring Strips

A biologic monitoring spore test strip is the *accepted weekly monitor of adequate time* and *temperature exposure*.

Spores dried on absorbent paper strips are calibrated to be killed when sterilization conditions are reached and maintained for the time necessary to kill all pathogenic microorganisms. By sending the strip to a licensed reference laboratory for testing, the dentist obtains *independent documentation* of monitoring frequency and sterilization effectiveness. In the event of failure, such laboratory personnel provide immediate expert consultation to help resolve the problem.

V. Documentation Log

In a log, a single, dated, initialled indicator strip is attached to a sheet or calendar for each workday, followed

by a weekly spore strip report. The log provides valuable sterilization documentation.

Dental Control Unit Water Systems

Oral fluid contamination problems of rotary equipment, especially the high-speed hand-piece, involve

I. contamination of hand-piece external surfaces and crevices,
II. turbine chamber contamination that enters the mouth,
III. water spray retraction and aspiration of oral fluids into the water lines of older dental units,
IV. growth of environmental aquatic bacteria in water lines and
V. exposure of personnel to spatter and aerosols generated by intraoral use of rotary equipment.[15,19,35,36]

Microbes exist in the dental unit water line as free-floating bacteria and as a sessile form known as biofilm. The microorganisms in the biofilm produce a protective polysaccharide matrix that provides them a mechanism for surface attachment and retention to the water line.[37] This matrix, which can be 30–50 microns thick, affords the biofilm flora resistance to antimicrobial agents on the order of 1500 times greater than normal free-floating bacteria. Because of this resistance to antimicrobial agents, when the biofilm is established, it is difficult to remove.

Clinical Notes

- Bacterial growth in biofilms on the inner walls of dental unit water lines (Fig. 4.11) is a universal occurrence unless steps are taken to control it.[37]
- Counts of bacteria that are shed from the biofilms into water of the dental unit may range from thousands to hundreds of thousands of bacteria per millilitre.[35,38–40]
- The main inhabitants are *opportunistic, gram-negative, aquaphilic bacteria.*
- Flushing or sterilizing high-speed hand-pieces cannot be expected to overcome this potential source of contamination of patients and personnel that extends throughout the dental unit water system.
- It is recommended that dental unit treatment water contain less than *500 cfu per millilitre* of bacteria.[1]
- Disinfectants such as an *iodophor* or *diluted sodium hypochlorite* are recommended to clean the system. These disinfectants must be flushed out with clean, boiled or sterile water before using the system.

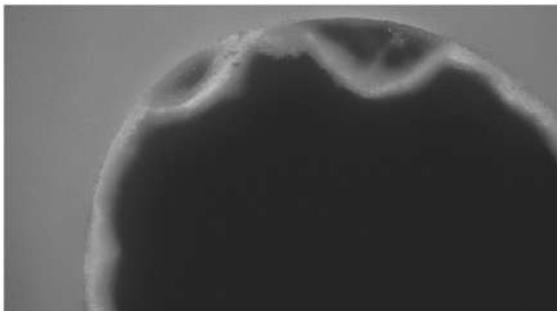


Figure 4.11 Magnification of cross-section of biofilm formation in dental unit waterline.
(From Bird DL, Robinson DS: *Modern Dental Assisting*, ed 10, St. Louis, MO, 2012, Saunders; Courtesy: Dr Shannon Mills.)

Infection Control for Impressions and Related Registrations Factors in Making Impressions and Associated Registrations to be Sent to a Remote Laboratory

- Precautions are required for infection control in making impressions and associated bite registrations.
- Universal barrier protection for personnel against contamination from mucosa, saliva and blood by use of adequate PPE such as gloves, mask and appropriate overgarment should be ensured. Before making the impression and associated bite registrations, clean, gloved hands should be used to dispense as many materials and disposable items as possible.
- Wiping material containers with a disinfectant after the procedures is the least satisfactory, but adequate, measure. An EPA-approved tuberculocidal disinfectant is considered satisfactory.
- For infection control, custom resin trays for impressions made with nonaqueous rubber impression materials are used once and then discarded. Likewise, stock trays are used only once and discarded. The tray size is indicated on the patient's chart to eliminate further try-ins.

Concepts for Transporting Impressions and Associated Registrations to a Remote Laboratory

Potentially infectious materials shall be placed in a container which prevents leakage during collection, handling, processing, storage, transport or shipping (to the laboratory). Labelling or colour coding is required when such specimens/containers leave the facility.[41]

Controversy exists over the two choices that may be used for preparing a potentially infectious item for transport:

I. Send it well cleaned (rinsed) and un-disinfected in a biohazard-labelled, heat-sealed, plastic bag or

II. debride, clean (rinse) and adequately disinfect it, place it in a sealed transport bag labelled with the precautions taken, and assume responsibility for the aseptic condition of the item.

In either case, most laboratories disinfect the item (a second time in the second choice) to ensure protection of laboratory personnel. Disinfecting twice wastes time, and multiple exposures to disinfectant should be avoided. The National Association of Dental Laboratories recommends disinfecting all items received from the dental office and disinfecting all appliances before shipping them from the laboratory.[41]

Inexpensive, biohazard-labelled, heat-sealable bags are commercially available in various sizes made of sturdy clear plastic and they are stamped with warnings to transporters and personnel.

Summary

It is not possible to provide all the details on barriers, standard operating procedures, sterilization methods and equipment in one chapter. The students are referred to other, more detailed literature and texts provided in the reference list and are advised to attend continuing education programs to expand and update their infection control information.[7,16,23–25,27,42]

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